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1. Introduction

The Software Requirements Specification (SRS) document provides a detailed and structured description of the **Building Disaster Risk Assessment and Safety Guidance Web Application**. This document defines the system's purpose, scope, functionality, constraints, and requirements in a clear and precise manner. It serves as a formal reference for stakeholders including project guides, developers, testers, and future maintainers.

Many residential and commercial buildings are vulnerable to various disasters such as earthquakes, fire accidents, floods, and extreme weather conditions. However, building occupants and owners often lack awareness about the actual risk level of their buildings and the safety facilities available within them. Existing disaster management solutions mainly focus on post-disaster response and alerts, rather than **pre-disaster risk assessment at the building level**.

The proposed Building Disaster Risk Assessment and Safety Guidance Web Application aims to address this gap by providing a practical and accessible platform that evaluates disaster risks based on building characteristics, location factors, and available safety measures. The system also offers safety precautions and emergency guidance tailored to specific disaster scenarios.

This SRS document is prepared in accordance with IEEE standards to ensure clarity, consistency, and completeness. The structured format enables both technical and non-technical readers to easily understand the system requirements and objectives.

1.1 Purpose

The purpose of this Software Requirements Specification document is to describe the functional and non-functional requirements of the **Building Disaster Risk Assessment and Safety Guidance Web Application** in detail. It defines how the system should behave, the constraints under which it operates, and the expected performance standards.

1.2 Document Conventions

This document follows standard IEEE SRS documentation conventions. Mandatory requirements are specified using the keyword “**shall**”, while recommended or optional requirements are described using “**should**”. Section headings and subheadings are numbered for easy navigation. Tables and lists are used where appropriate to enhance clarity and readability.

1.3 Project Scope

The Building Disaster Risk Assessment and Safety Guidance Web Application is a **web-based system** designed to assess the disaster risk level of buildings for selected disaster types such as earthquakes, fire accidents, floods, and extreme weather events.

1.4 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirements Specification
UI	User Interface
GIS	Geographic Information System
API	Application Programming Interface
Risk Level	Safe, Moderate Risk, or High Risk

2. Overall Description

2.1 Product Perspective

The Building Disaster Risk Assessment and Safety Guidance System is a standalone web-based application designed to evaluate disaster risks at the building level. The system can be integrated with government portals, smart city platforms, or institutional safety systems in the future. It uses a centralized database to store building information, disaster risk parameters, and safety facility details in a structured and secure manner.

Users interact with the system through a web browser, eliminating the need for additional software installation. The application provides rule-based risk assessment for selected disasters such as earthquakes, fire accidents, floods, and extreme weather conditions. This approach improves disaster preparedness, reduces dependency on manual assessments, and supports scalability for multiple locations and buildings.

2.2 Product Functions

The major functions of the system include building registration, disaster risk assessment, safety facility evaluation, disaster-specific precaution guidance, and identification of nearby emergency support centers. The system analyzes user-provided building data and predefined risk rules to classify risk levels.

Each function is designed to increase awareness and preparedness rather than real-time disaster prediction. The system ensures consistent assessment results and provides clear safety guidance for different disaster scenarios. These functions work together to support informed decision-making regarding building safety.

2.3 Operating Environment

The system operates in a web-based environment and can be accessed through modern web browsers such as Chrome, Firefox, and Edge. No platform-specific installation is required. The backend application runs on servers that support standard database management systems.

The system is compatible with commonly used operating systems such as Windows and Linux, ensuring wide accessibility across different devices and locations.

2.4 Design and Implementation Constraints

The system must be developed using secure authentication mechanisms to protect user and building data. It must follow standard web application development practices and comply with basic data security guidelines. Database-driven architecture shall be used to ensure efficient storage and retrieval of information.

Performance constraints require the system to respond within acceptable time limits for user interactions. The system does not support real-time disaster alerts or structural engineering simulations, which are considered outside the project scope.

2.5 Assumptions and Dependencies

The system assumes reliable internet connectivity for proper operation. Availability of server infrastructure is required to host the application. The system depends on a database service for storing building details, disaster parameters, and safety facility information.

Risk assessment accuracy depends on the correctness of user-provided data and predefined disaster rules. Changes in disaster guidelines or regulations may require updates to the system logic.

3. External Interface Requirements

The External Interface Requirements describe how the Building Disaster Risk Assessment and Safety Guidance System interacts with users, software components, and communication mechanisms. These interfaces define system boundaries and ensure effective interaction, usability, and data exchange.

3.1 User Interfaces

The system provides a user-friendly web interface designed for both administrators and users. The interface is browser-based and accessible without additional software installation. It is designed with clarity, simplicity, and ease of navigation to support users with minimal technical knowledge.

3.2 Hardware Interfaces

The system primarily operates as a software-based application and does not require specialized hardware. Optional location services such as GPS (through browser-based location access) may be used to improve flood risk assessment and emergency center identification.

3.3 Software Interfaces

The system interacts with a relational Database Management System such as MySQL to store building details, risk assessment rules, safety facility data, and user information. Backend services handle create, read, update, and delete operations securely.

A web server hosts the application and manages requests from client browsers. Integration between the frontend, backend, and database ensures reliable data handling and consistent system behavior.

3.4 Communication Interfaces

The system uses standard HTTP and HTTPS protocols for communication between client devices and the server. HTTPS is used to secure sensitive data such as login credentials and building information.

The system follows a REST-based communication approach between frontend and backend components. Data exchange occurs in structured formats, supporting scalability and maintainability.

4. System Features and Requirements

4.1 Functional Requirements

1. FR-1.1 The system shall allow users to input building details such as type, age, number of floors, and construction material.
2. FR-1.2 The system shall capture and validate user location data.
3. FR-1.3 The system shall support both automatic (GPS-based) and manual location detection.
4. FR-2.1 The system shall clean, normalize, and validate user input data.
5. FR-2.2 The system shall integrate external disaster risk datasets such as flood zones, seismic maps, and cyclone-prone regions.
6. FR-3.1 The system shall calculate a risk score using structural, environmental, and locational factors.
7. FR-3.2 The system shall classify buildings as Safe, Moderate Risk, or High Risk.
8. FR-3.3 The system shall support analysis for earthquakes, floods, fires, and cyclones.
9. FR-4.1 The system shall generate personalized safety instructions based on the predicted risk level.
10. FR-4.2 The system shall recommend nearby safe places such as shelters, hospitals, and evacuation areas.
11. FR-4.3 The system shall provide guidance during emergency situations.
12. FR-5.1 The system shall provide a simple and intuitive user interface.

13. FR-5.2 The system shall display results using textual and visual formats such as maps and icons.

5. Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system. These requirements focus on how the system performs rather than the specific functions it provides. They ensure that the system is efficient, secure, reliable, and user-friendly under expected operating conditions. Non-functional requirements play a critical role in improving system usability, dependability, and long-term sustainability.

5.1 Performance Requirements

- The system shall load web pages within **2 seconds** under normal operating conditions.
- The system shall provide fast response times for user interactions such as risk assessment and report generation.
- The system shall handle multiple concurrent users efficiently.
- Performance consistency shall be maintained during regular system usage.
- These requirements ensure smooth and responsive system operation.

5.2 Security Requirements

- User passwords shall be encrypted to protect authentication credentials.
- Secure authentication mechanisms shall be implemented to prevent unauthorized access.
- User and building data shall be protected against data breaches and misuse.
- Role-based access control shall be enforced for Admin and User roles.
- All sensitive data transmission shall use secure communication protocols.

5.3 Reliability and Availability

- The system shall be available **99% of the time** under normal operating conditions.
- High availability ensures continuous access for users during critical situations.
- System downtime shall be minimized through stable hosting infrastructure.
- Reliable system behavior builds user trust and confidence.

5.4 Maintainability and Scalability

- The system shall be modular to support easy maintenance and updates.
- New disaster types or safety rules should be added without affecting existing functionality.
- The system architecture shall support scalability to accommodate increased users and buildings.
- Clear documentation shall support future system enhancements and maintenance.

5.5 Legal and Regulatory Requirements

- User data shall be handled securely and confidentially.
- Basic data privacy guidelines shall be followed strictly.
- Sensitive information such as location and building details shall be protected.
- Data handling practices shall comply with institutional and ethical standards.

6. External Interface Requirements

The External Interface Requirements describe how the system interacts with users, software components, and communication services. These interfaces define system boundaries and ensure smooth interaction and data exchange.

6.1 User Interfaces

- Responsive web portal accessible through standard web browsers
- Administrator dashboard for system and data management
- User dashboard for building risk assessment and safety guidance
- Simple and intuitive interface for non-technical users

6.2 Hardware Interfaces

- Optional GPS module for location-based flood risk assessment
- Optional device camera access for future enhancements
- No mandatory specialized hardware required

6.3 Software Interfaces

- Relational Database Management System (e.g., MySQL)
- Web server to host the application
- Integration with GIS APIs for location-based risk reference
- Optional integration with government disaster-related datasets

6.4 Communication Interfaces

- HTTPS protocol for secure communication
- REST-based communication between frontend and backend
- Structured data exchange for scalability and maintainability

7. System Architecture and Module Description

The system follows a modular architecture to ensure separation of concerns, ease of maintenance, and scalability.

Module 1: User Input and Data Collection Module

This module collects and validates building details, location information, and safety facility data provided by users.

Module 2: Data Preprocessing and Integration Module

This module cleans and organizes input data and integrates predefined disaster risk parameters and datasets.

Module 3: Disaster Risk Analysis and Building Safety Module

This module evaluates disaster risks based on building characteristics and classifies risk levels such as Low, Moderate, or High.

Module 4: Safety Guidelines and Decision Support Module

This module provides disaster-specific safety instructions, precautionary measures, and emergency support center recommendations.

8. Appendices

8.1 Analysis Models

8.1.1 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates the flow of information within the Building Disaster Risk Assessment and Safety Guidance System. External entities include the **User** and **Administrator**. Inputs such as building details and safety data are processed by the system and stored in the database. Outputs include risk levels, safety recommendations, and emergency center information.

Level 0 DFD represents the system as a single process. Level 1 DFD decomposes it into processes such as Data Collection, Risk Assessment, Safety Guidance, and Emergency Center Identification.

8.1.2 Entity Relationship Diagram (ERD)

The ERD defines the database structure of the system. Key entities include User, Building, DisasterType, RiskAssessment, SafetyFacility, and EmergencyCenter. Relationships represent how buildings are assessed for disaster risks and how safety recommendations are linked.

8.1.3 State Transition Diagram

State transition diagrams represent changes in building risk status such as Low Risk, Moderate Risk, and High Risk based on assessment results and updated data.

8.2 Detailed Use Cases

Use Case 1: User Login

- **Actor:** User/Admin
- **Description:** User logs into the system using valid credentials
- **Precondition:** User must be registered
- **Postcondition:** User is redirected to the appropriate dashboard

Use Case 2: Register Building

- **Actor:** User
- **Description:** User enters building and safety details
- **Outcome:** Building data is stored successfully

Use Case 3: Assess Disaster Risk

- **Actor:** User
- **Description:** User requests disaster risk assessment
- **Outcome:** System displays risk level and recommendations

Use Case 4: View Safety Guidelines

- **Actor:** User
- **Description:** User views disaster-specific safety precautions
- **Outcome:** Safety instructions are displayed

Use Case 5: Manage Disaster Parameters

- **Actor:** Admin
- **Description:** Admin updates disaster rules and safety criteria

8.3 Requirements Traceability Matrix (RTM)

The RTM ensures that each requirement is traced through design, implementation, and testing. It maps functional requirements to corresponding use cases and test cases to ensure full coverage.

9. Future Enhancements

- Mobile application support (Android and iOS)
- Real-time disaster alert integration
- AI-based risk prediction models
- Integration with official government disaster warning systems
- Advanced visualization using maps and dashboards

10. Conclusion

The Building Disaster Risk Assessment and Safety Guidance Web Application provides a practical and reliable solution for assessing building-level disaster risks and improving preparedness. By offering structured risk evaluation and clear safety guidance, the system enhances awareness and supports informed decision-making. This SRS document serves as a comprehensive reference for system development, testing, and future enhancement.

11. References

- Project Proposal Document: ResQGuide
- NDMA Disaster Management Guidelines.
- <https://cdnbbsr.s3waas.gov.in/s3kv027b571733cd04d2b1611257c05049/uploads/2024/05/2024053090.pdf>